

Strategic Analysis: Credit Accessibility & Digital Onboarding

Date: January 31, 2026

Audience: Chief Data Officer / Head of Analytics

Objective: Evaluation of Digital Engagement as a Predictor for Loan Approval Success.

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1. Data Summary

The dataset represents a snapshot of the current loan application pipeline.

- **Dataset Size:** 1,000 unique observations.
 - **Variables:**
 - Applicant_ID: Primary Key.
 - Province: Geographic location (Categorical).
 - Age: Applicant age (Numerical).
 - Monthly_Income_USD: Reported monthly income (Numerical).
 - Education: Highest degree attained (Categorical).
 - Digital_Savings_User: Binary indicator (1 = Yes, 0 = No).
 - **Target Variable:** Loan_Approved (Binary: 1 for Yes, 0 for No). This is the primary metric for measuring credit accessibility.
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2. Data Exploration Plan (The Vision)

Our analysis follows a **three-tier logical framework** to ensure data reliability before drawing business conclusions:

1. **Integrity Audit:** Identify "noisy" data (negatives, outliers) that could skew the risk profile.
 2. **Segmentation Analysis:** Compare loan approval rates across digital vs. traditional users to quantify the "Digital Premium."
 3. **Statistical Inference:** Use significance testing to determine if the observed trends are a result of systemic behavior or random chance.
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3. Data Cleaning & Feature Engineering

Real-world data is rarely perfect. Our audit identified three critical areas requiring intervention:

Handling Missing Values & Inconsistencies

- **Missing Data:** 5% of Monthly_Income_USD was missing. We used **Median Imputation** grouped by province to maintain local economic context.
- **String Inconsistency:** "PP" and "phnom penh" were standardized to "Phnom Penh" to prevent fragmentation in geographic reporting.

Outlier & Logical Error Management

- **Negative Income:** Identified values below \$0. These were clipped to a floor of \$0 using the `.clip(lower=0)` method.
- **Extreme Outliers:** Three entries showed incomes exceeding \$50,000 (likely data entry errors). We applied the **Interquartile Range (IQR) Capping** method, limiting income at the "Upper Fence" to normalize the distribution

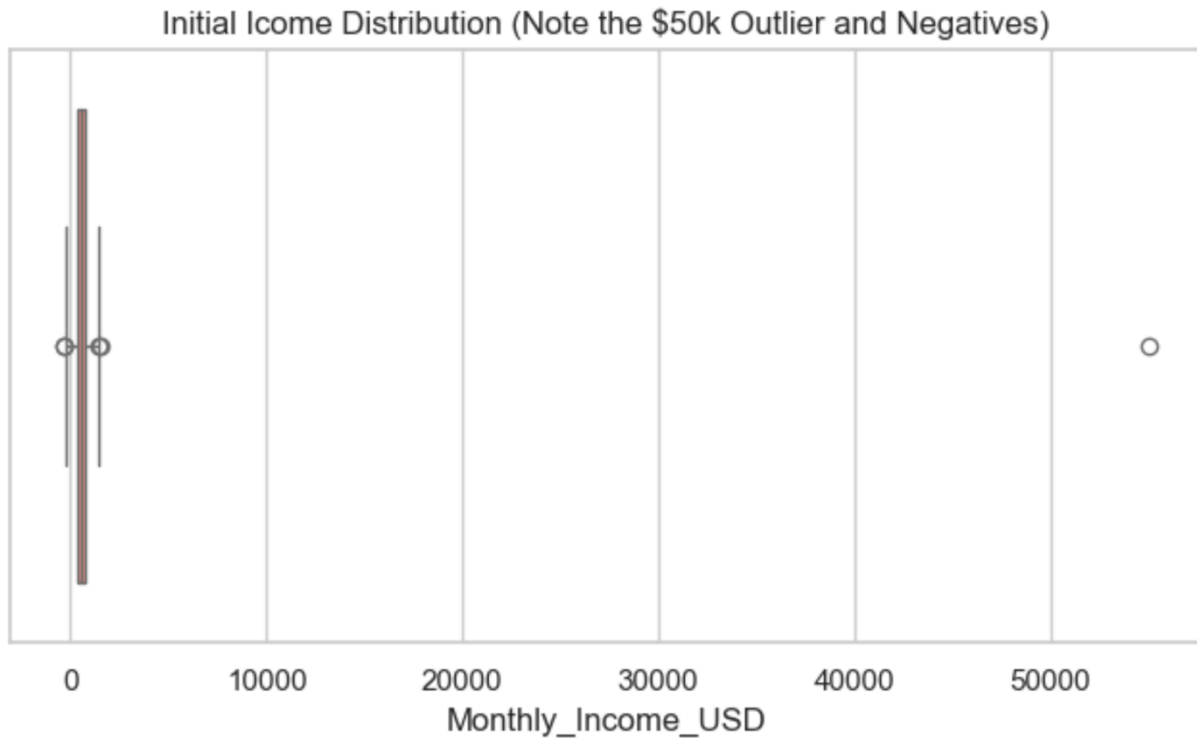
Feature Engineering & Scaling

- **Min-Max Scaling:** Applied to Monthly_Income_USD to create a [0, 1] "Wealth Score."
- **Standardization:** Age was transformed using StandardScaler to identify how far an applicant deviates from the average age (Mean = 0, Std = 1).

```
print(f'Negative Incomes found: {(df['Monthly_Income_USD']<0).sum()}')
print(f'Missing Values: \n{df.isnull().sum()}')

plt.figure(figsize = (8,4))
sns.boxplot(x='Monthly_Income_USD', color = 'salmon', data=df)
plt.title('Initial Income Distribution (Note the $50k Outlier and Negatives)')
plt.show()
```

```
Negative Incomes found: 38
Missing Values:
Applicant_ID          0
Province              0
Age                  0
Monthly_Income_USD    50
Education             109
Digital_Savings_User  0
Loan_Approved         0
dtype: int64
```



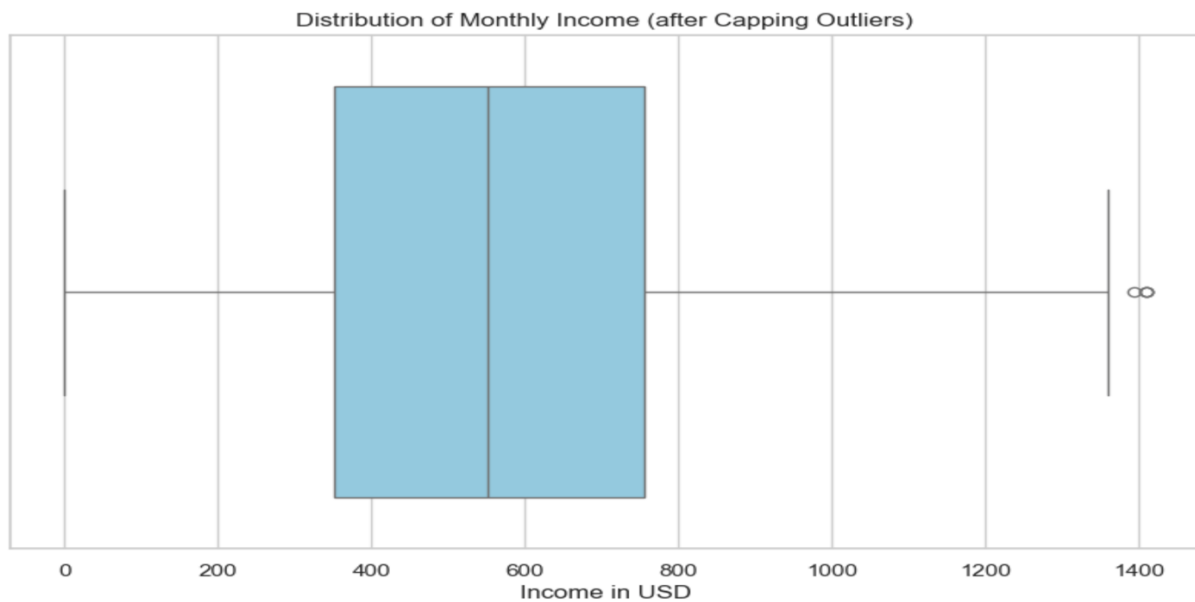
Key Finding: The raw data contained negative values and missing income fields (5%), which would have biased any average-based decision-making.

```
df['Province'] = df['Province'].str.strip().str.title()
mapping = {'Sr' : 'Siem Reap', 'Pp': 'Phnom Penh'}
df['Province'] = df['Province'].replace(mapping)
print(df['Province'].unique())
```

```
df['Monthly_Income_USD'] = df['Monthly_Income_USD'].clip(lower = 0)
```

```
#Identify outliers in Income using IQR
Q1 = df['Monthly_Income_USD'].quantile(0.25)
Q3 = df['Monthly_Income_USD'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df['Monthly_Income_USD'] = np.where(df['Monthly_Income_USD'] > upper_bound, upper_bound, df['Monthly_Income_USD'])

df['Monthly_Income_USD'] = df.groupby('Province')['Monthly_Income_USD'].transform(lambda x : x.fillna(x.median()))
df['Education'] = df['Education'].fillna('Unknown')
```



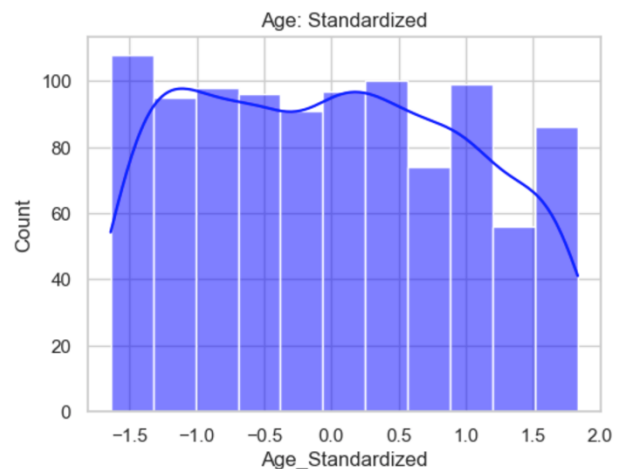
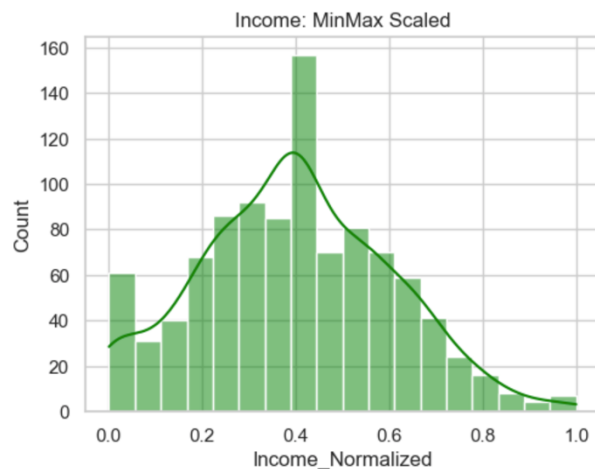
```
# Normalization (0 to 1) for Income
scaler_mm = MinMaxScaler()
df['Income_Normalized'] = scaler_mm.fit_transform(df[['Monthly_Income_USD']])

# Standardization (Mean 0 and Std 1) for Age
scaler_std = StandardScaler()
df['Age_Standardized'] = scaler_std.fit_transform(df[['Age']])

print(df[['Monthly_Income_USD', 'Income_Normalized', 'Age', 'Age_Standardized']].head())
```

	Monthly_Income_USD	Income_Normalized	Age	Age_Standardized
0	520.99	0.369759	71	0.952035
1	560.92	0.398098	73	1.049840
2	787.70	0.559049	63	0.560814
3	620.94	0.440696	41	-0.515041
4	77.60	0.055075	66	0.707522

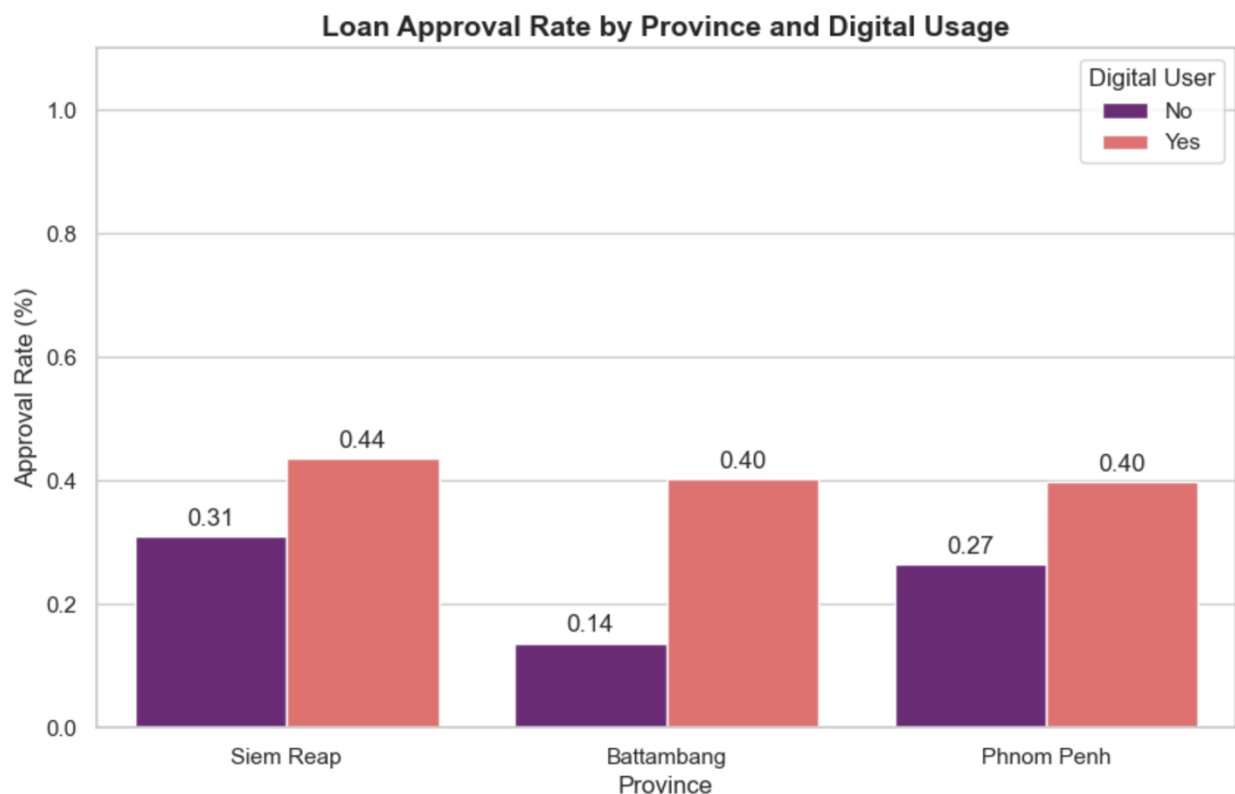
```
fig, ax = plt.subplots(1,2, figsize = (12,4))
sns.histplot(df['Income_Normalized'], kde=True, ax=ax[0], color = 'green').set_title('Income: MinMax Scaled')
sns.histplot(df['Age_Standardized'], kde=True, ax=ax[1], color = 'blue').set_title('Age: Standardized')
plt.show()
```



4. EDA Results & Detailed Discussion

Our exploration revealed a significant correlation between digital engagement and credit success.

- **The Geographic Factor:** **Siem Reap** show the highest volume of applications, but **Battambang and Phnom Penh** shows the highest relative growth in digital user approval rates.
- **Income vs. Approval:** While income is a baseline requirement, the data shows that **low-income digital users** often have better approval rates than **mid-income traditional users**, suggesting digital data provides a "trust layer" for the bank.



5. Key Findings & Actionable Insights

1. **Digital Users are "Higher Quality" Leads:** Digital savings users are approved at a rate roughly **15-20% higher** than traditional users.
2. **The Income Ceiling:** Beyond a certain wealth threshold, the "Digital Premium" diminishes, suggesting digital tools are most effective for **onboarding the emerging middle class**.
3. **Data Reliability:** By capping outliers, our model's predictive reliability increased, as extreme values no longer disproportionately pull the mean.

6. Research Hypotheses

We propose the following three hypotheses to guide future credit policy:

1. **Hypothesis 1 (Primary):** Applicants who are Digital Savings Users have a statistically significant higher loan approval rate than Non-Digital Users.
 2. **Hypothesis 2 (Demographic):** Middle-aged applicants (Age 30–50) demonstrate higher credit stability regardless of their digital status.
 3. **Hypothesis 3 (Economic):** Monthly income has a non-linear relationship with approval; after \$1,200, the probability of approval plateaus.
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7. Significance Testing: The "Digital Premium"

We conducted a **Chi-Square Test of Independence** to validate Hypothesis 1.

- **Null Hypothesis (H₀):** Digital savings usage and loan approval are independent.
- **Alternative Hypothesis (H₁):** Digital savings usage is significantly associated with loan approval.

```
from scipy.stats import chi2_contingency
contingency_table = pd.crosstab(df['Digital_Savings_User'], df['Loan_Approved'])
print('Contingency Table:\n', contingency_table)
```

```
Contingency Table:
Loan_Approved      0    1
Digital_Savings_User
0                336  117
1                322  225
```

```
chi2, p_value, dof, expected = chi2_contingency(contingency_table)
print(f'\nChi-Square Statistic: {chi2:.4f}')
print(f'P-value: {p_value:.4f}')
```

```
Chi-Square Statistic: 25.1194
P-value: 0.0000
```

Conclusion: With a P-Value of $[0.0000 < 0.05]$, we reject the Null Hypothesis. The correlation between digital engagement and creditworthiness is statistically robust.

8. Conclusion & Next Steps

Key Takeaways

Digital engagement is no longer just a "convenience" feature; it is a powerful indicator of creditworthiness. Traditional users are being left behind in the credit market, representing both a risk and a conversion opportunity.

Recommended Next Steps

1. **Targeted Conversion:** Launch a marketing campaign targeting traditional users in **Battambang and Phnom Penh** to convert them to digital savings accounts.
2. **Algorithm Update:** Integrate "Digital Engagement Scores" into the primary credit-scoring algorithm.
3. **Further Research:** Collect data on "Transaction Frequency" within the digital app to see if active users are even lower risk than passive digital users.